



Discover data format

I don't know about you, but when I'm choosing a movie to watch, I sometimes get stuck between a couple of choices. If I'm in the mood for excitement or suspense, I might go for a thriller, but if I need a good laugh, I'll choose a comedy. If I really can't decide between two movies, I might even use some of my data analysis skills to compare and **contrast** them. Come to think of it, there really needs to be more movies about data analysts. I'd watch that, but since we can't watch a movie about data, at least not yet, we'll do the next best thing: watch data about movies! We're going to take a look at this spreadsheet with movie data. We know we can compare different movies and movie genres. Turns out, you can do the same with data and data formats. Let's use our movie data spreadsheet to understand how that works. We'll start with quantitative and qualitative data. If we check out column A, we'll find titles of the movies. This is **qualitative data** because **it can't be counted, measured, or easily expressed using numbers**. **Qualitative data is usually listed as a name, category, or description**. In our spreadsheet, the movie titles and cast members are qualitative data. Next up is **quantitative data, which can be measured or counted and then expressed as a number**. **This is data with a certain quantity, amount, or range**. In our spreadsheet here, the last two columns show the movies's budget and box office revenue. The data in these columns is listed in dollars, which can be counted, so we know that data is quantitative.

We can go even deeper into quantitative data and break it down into discrete or continuous data. Let's check out discrete data first. This is data that's counted and has a limited number of values. Going back to our spreadsheet, we'll find each movie's budget and box office returns in columns M and N. These are both examples of discrete data that can be counted and have a limited number of values. For example, the amount of money a movie makes can only be represented with exactly two digits after the decimal to represent cents. There can't be anything between one and two cents.

Reflect

Discrete data isn't limited to dollar amounts. Examples of other discrete data are stars and points. When partial measurements (half-stars or quarter-points) aren't allowed, the data is discrete. If you don't accept anything other than full stars or points, the data is considered discrete.

Continuous data can be measured using a timer, and its value can be shown as a decimal with several places. Let's imagine a movie about data analysts that I'm definitely going to star in someday. You could express that movie's run time as 110.0356 minutes. You could even add fractional data after the decimal point if you needed to.

There's also nominal and ordinal data. Nominal data is a type of qualitative data that's categorized without a set order. In other words, this data doesn't have a sequence.

Here's a quick example. Let's say you're collecting data about movies. You ask people if they've watched a given movie. Their responses would be in the form of nominal data.

They could respond "Yes," "No," or "Not sure." These choices don't have a particular

order. Ordinal data, on the other hand, is a type of qualitative data with a set order or

scale. If you asked a group of people to rank a movie from 1 to 5, some might rank it as

a 2, others a 4, and so on. These rankings are in order of how much each person liked the movie. Now let's talk about **internal data**, which is data that lives within a company's own systems. For example, if a movie studio had compiled all of the data in the spreadsheet using only their own collection methods, then it would be their internal data. The great thing about internal data is that it's usually more reliable and easier to collect, but in this spreadsheet, it's more likely that the movie studio had to use data owned or shared by other studios and sources because it includes movies they didn't make. That means they'd be collecting external data. **External data is**, you guessed it, **data that lives and is generated outside of an organization**. External data becomes particularly valuable when your analysis depends on as many sources as possible. A great thing about this **data is that it's structured**. **Structured data is data that's organized in a certain format, such as rows and columns**. Spreadsheets and relational databases are two examples of software that can store data in a structured way. **You might remember our earlier exploration of structured thinking, which helps you add a framework to a problem so that you can solve it in an organized and logical manner. You can think of structured data in the same way. Having a framework for the data makes the data easily searchable and more analysis-ready**. As a data analyst, you'll work with a lot of **structured data**, which will **usually be in the form of a table, spreadsheet or relational database**, but sometimes you'll come across **unstructured data**. **This is data that is not organized in any easily identifiable manner. Audio and video files are examples of unstructured data because there's no clear way to identify or organize their content**. Unstructured data might have internal structure, but the data doesn't fit neatly in rows and columns like structured data. And there you have it! Hopefully you're now

more familiar with data formats and how you might use them in your work. In just a bit, you'll continue to explore structured data and learn even more about the data you'll use most often as an analyst. Coming soon to a screen near you.

An entertainment website displays a star rating for a movie based on user reviews. Users can select from one to five whole stars to rate the movie. The star rating is an example of what type of data? Select all that apply.

☒ Ordinal

✔ **Correct**

The star rating is an example of ordinal data because the number of stars are in order of how much each person liked the movie. It's also an example of discrete data because a person has to choose a full star measure; half-stars weren't an option.

☒ Discrete

✔ **Correct**

The star rating is an example of ordinal data because the number of stars are in order of how much each person liked the movie. It's also an example of discrete data because a person has to choose a full star measure; half-stars weren't an option.

☐ Nominal

☐ Continuous
